



DRISHTI: Resolving the Post-Disaster Information Fog by Treating Silence as Evidence

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ABSTRACT

In the first 24 hours after a climate-driven multi-hazard event, a District Emergency Operations Centre (EOC) receives thousands of fragmented, contradictory and unverified ground reports, while severed roads prevent physical verification. Scarce rescue assets are therefore allocated to wherever reporting is loudest.

We identify a systematic failure in this practice: report volume is non-monotonic in damage. Undamaged settlements generate no reports because there is nothing to report, and catastrophically damaged settlements also generate none because casualties and destroyed telecom infrastructure remove the reporters themselves. Volume-based triage therefore ranks destroyed villages alongside safe ones.

DRISHTI reframes the task from establishing truth to maintaining a calibrated belief. Reports are geo-resolved to a probability distribution over settlements, parsed for claims, clustered so that amplified rumour cannot masquerade as corroboration, and fused by Dempster-Shafer combination of simple support functions in closed form. Critically, ABSENCE is converted into positive evidence by fusing telecom-tower liveness, chatter deficit against a population baseline, and terrain exposure.

Evaluated on a 1,102-settlement district built from OpenStreetMap with hidden ground truth, DRISHTI reaches 6,057 of 10,928 affected people within 24 h against 1,006 for volume ranking, using identical assets. For the 492 settlements that emit no reports at all, report-driven ranking is statistically indistinguishable from chance (AUC 0.500) whereas DRISHTI attains 0.848.

Keywords: disaster informatics, evidence fusion, Dempster-Shafer theory, climate resilience, humanitarian logistics, open geospatial data

KEY OUTCOMES

- 6.0x more affected people reached with identical assets
- AUC 0.848 vs 0.500 on settlements that never reported
- Blackout detection at 64% precision vs 34% base rate
- ~900 of 3,256 reports suppressed as panic amplification
- Runs offline on one laptop · zero licence or cloud cost

DISCUSSION

Why silence is hard. An undamaged village and a destroyed village are both silent, so report volume alone can never separate them. We resolve this with telecom-tower liveness, an observation about the PLACE rather than about whether anyone had something to say.

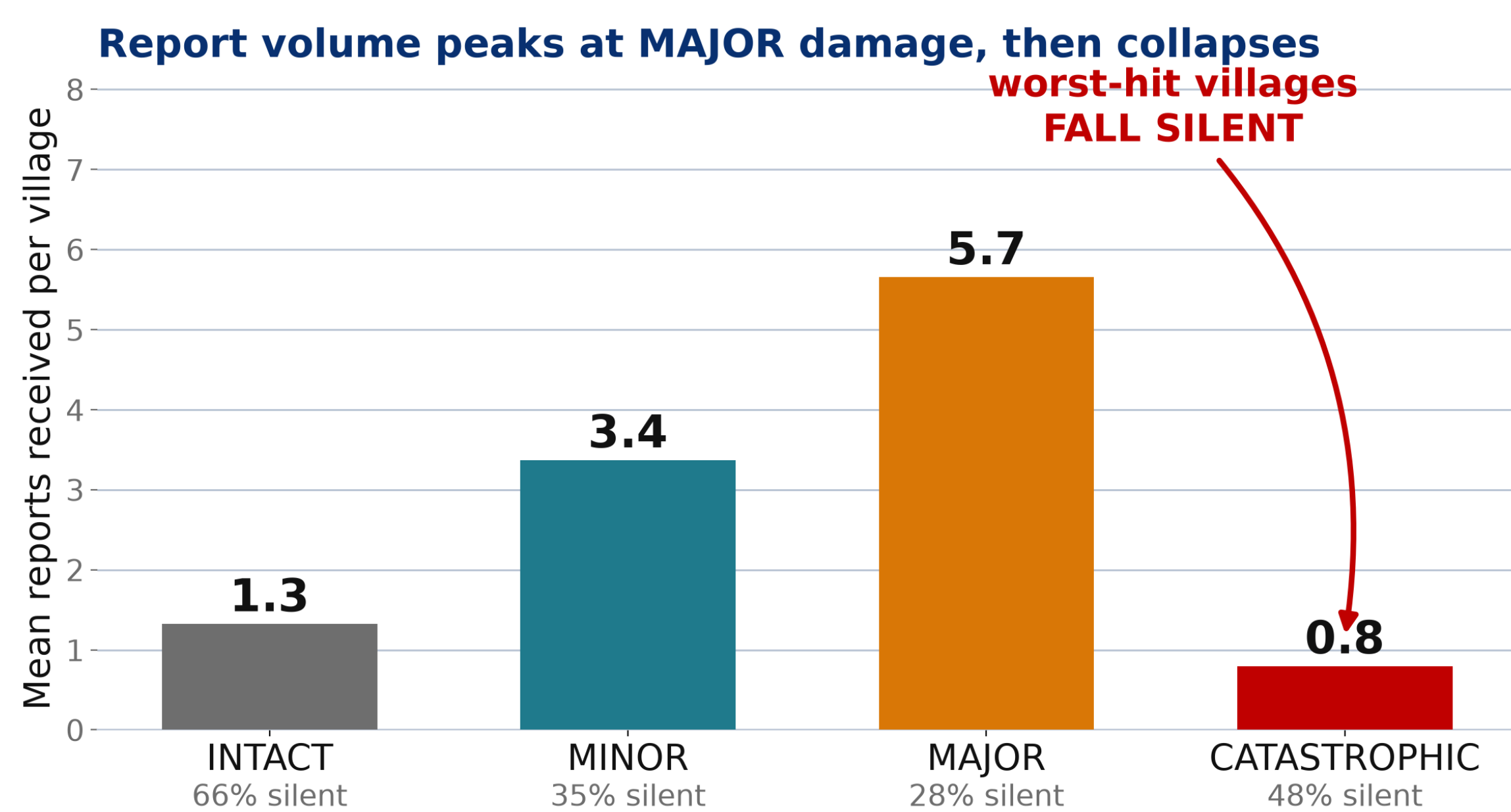
Tower loss is not proof. In our scenario 51% of undamaged settlements also lose their tower, because the grid is cut district-wide to prevent electrocution. Tower-down is therefore treated as weak evidence and never used alone; terrain exposure supplies the prior that separates structural collapse from a generator that ran dry.

Guarding against circular evaluation. An early version of our simulator generated damage from the same terrain variables the engine's prior consumes, yielding an inflated AUC of 0.98. We introduced a spatially-correlated latent vulnerability field (building stock, drainage, upstream release) that no free map layer reveals. The honest figure is 0.857.

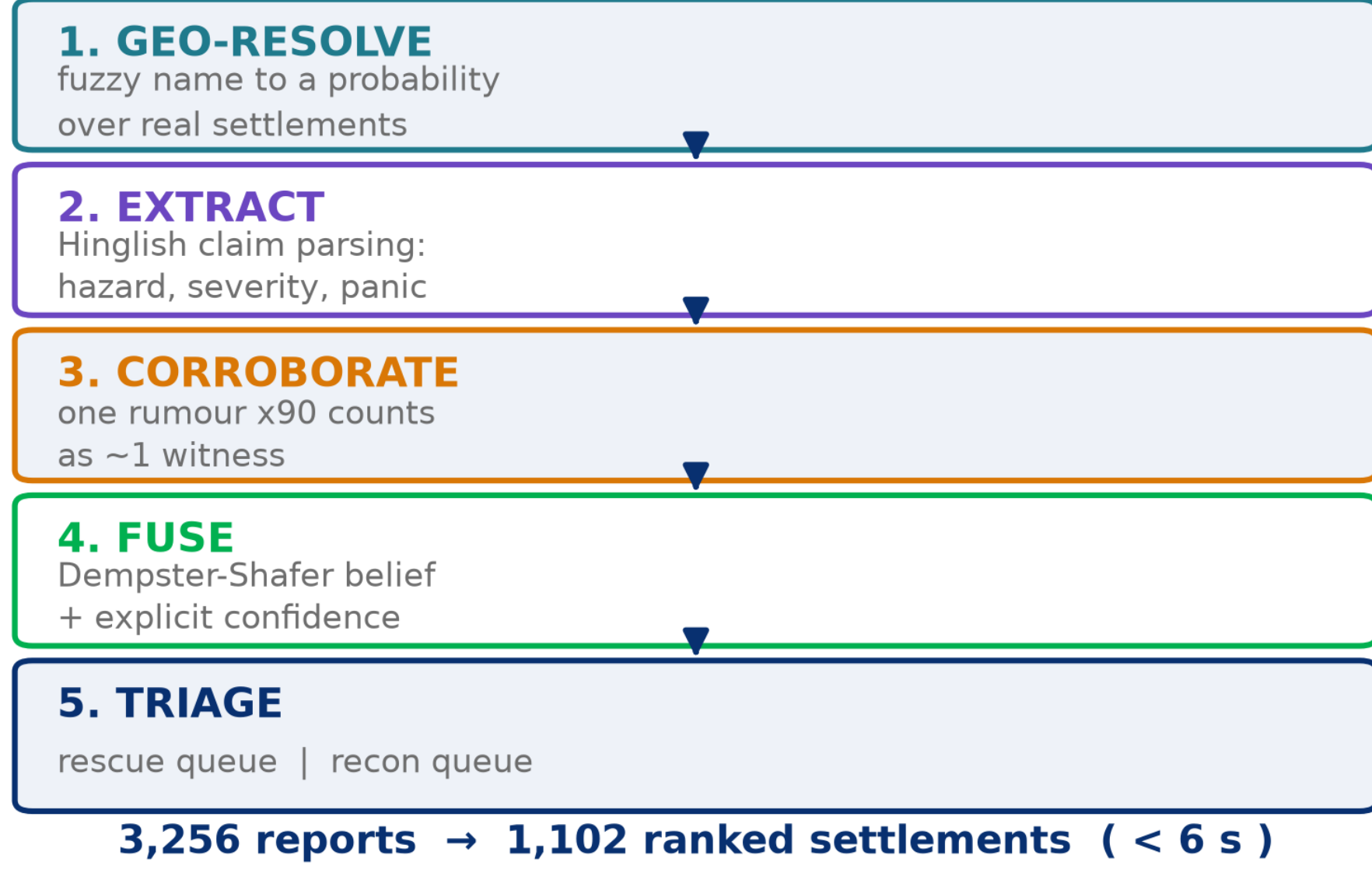
Ablation. Removing the silence engine costs 0.061 AUC overall and 0.137 on non-reporting settlements, confirming that absence carries genuine information rather than acting as a tie-breaker.

Limitations. Ground truth is simulated, since no labelled multi-hazard report corpus exists for an Indian district; population is estimated from OSM place class where census figures are absent; and the telecom feed is modelled rather than live. None of these affect the architecture, only the calibration constants.

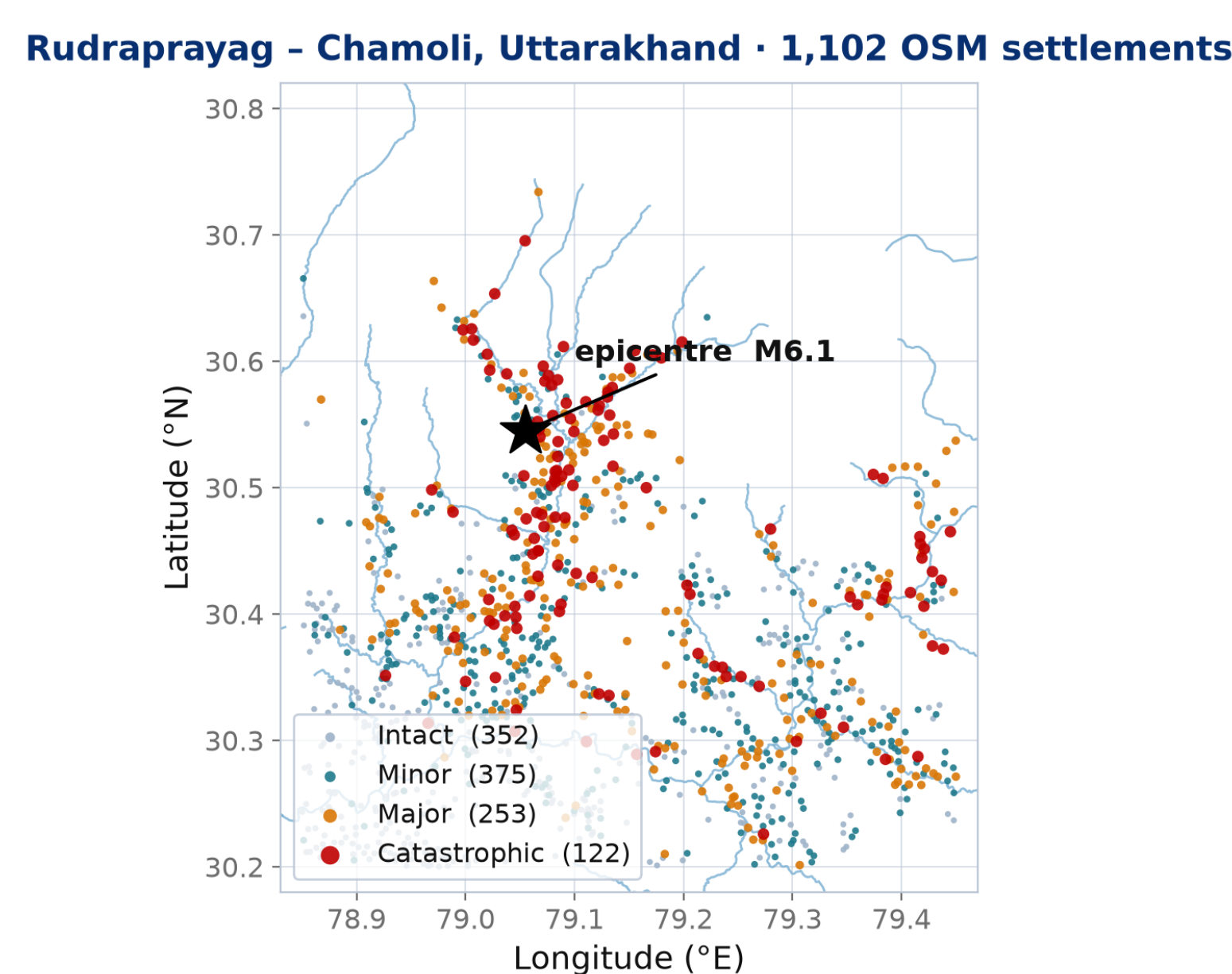
MOTIVATION



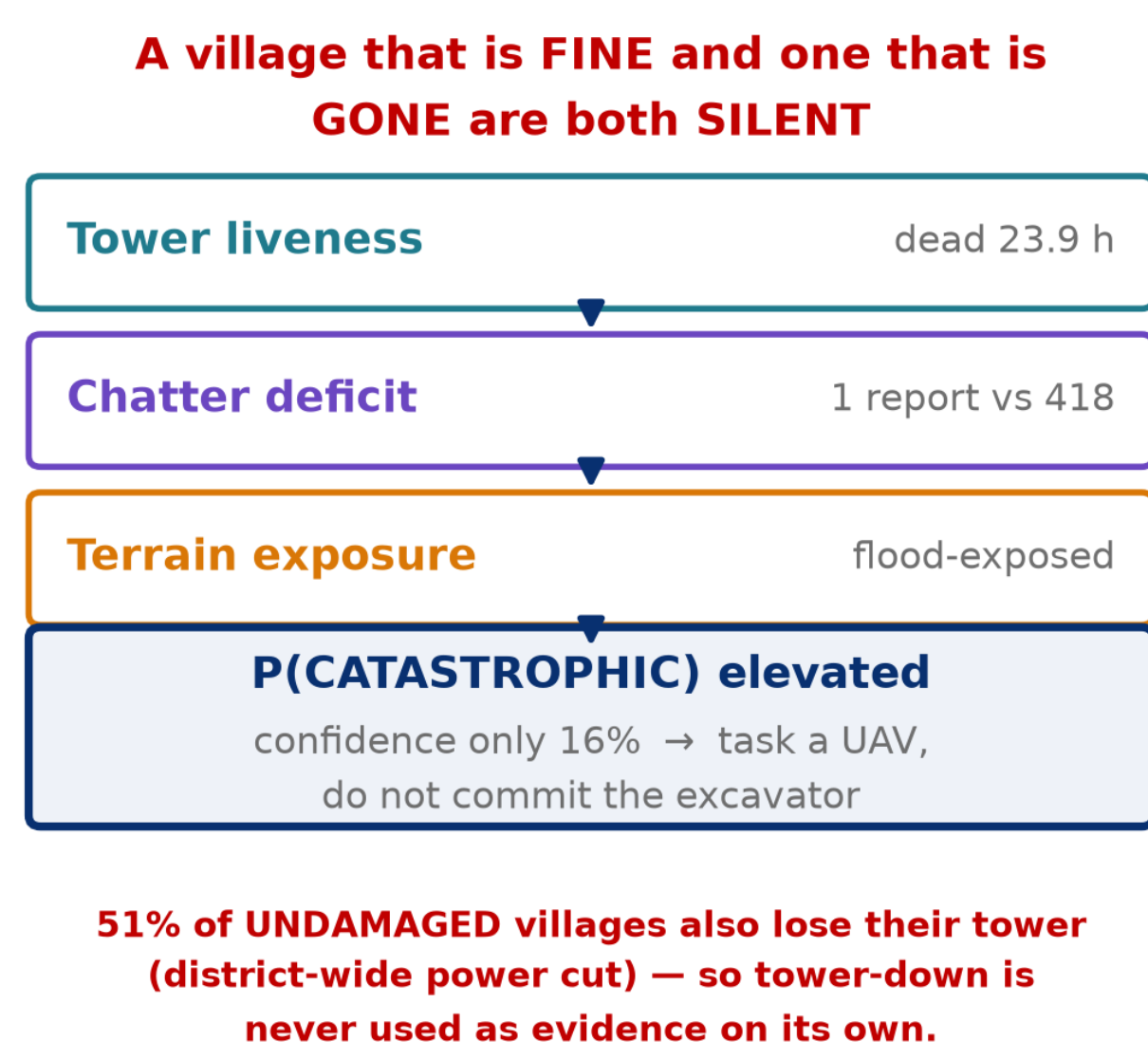
MATERIALS AND METHOD



EXPERIMENTAL SETUP



MECHANISM



RESULT

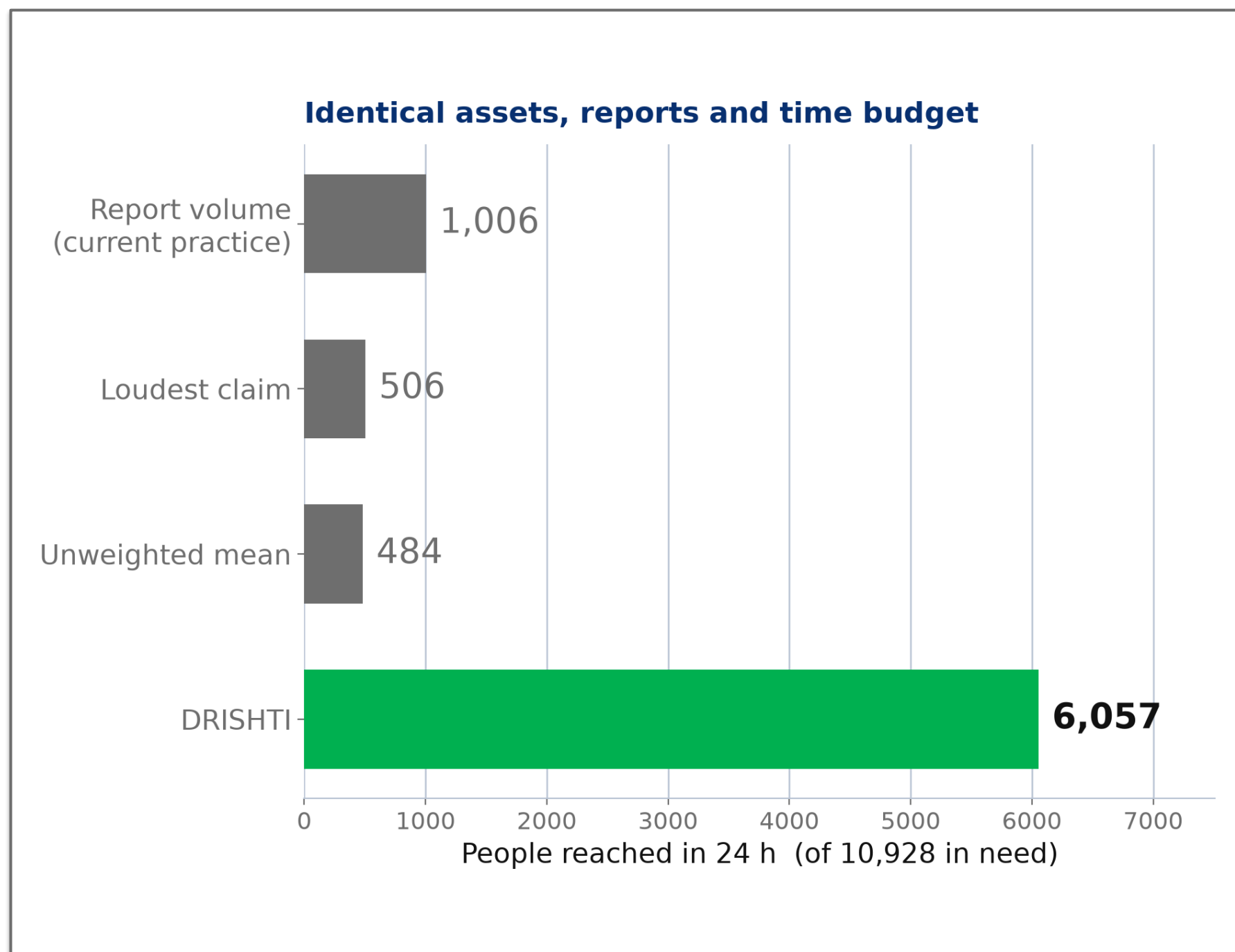


Fig. 1 People reached within 24 h under a fixed 40-asset budget.

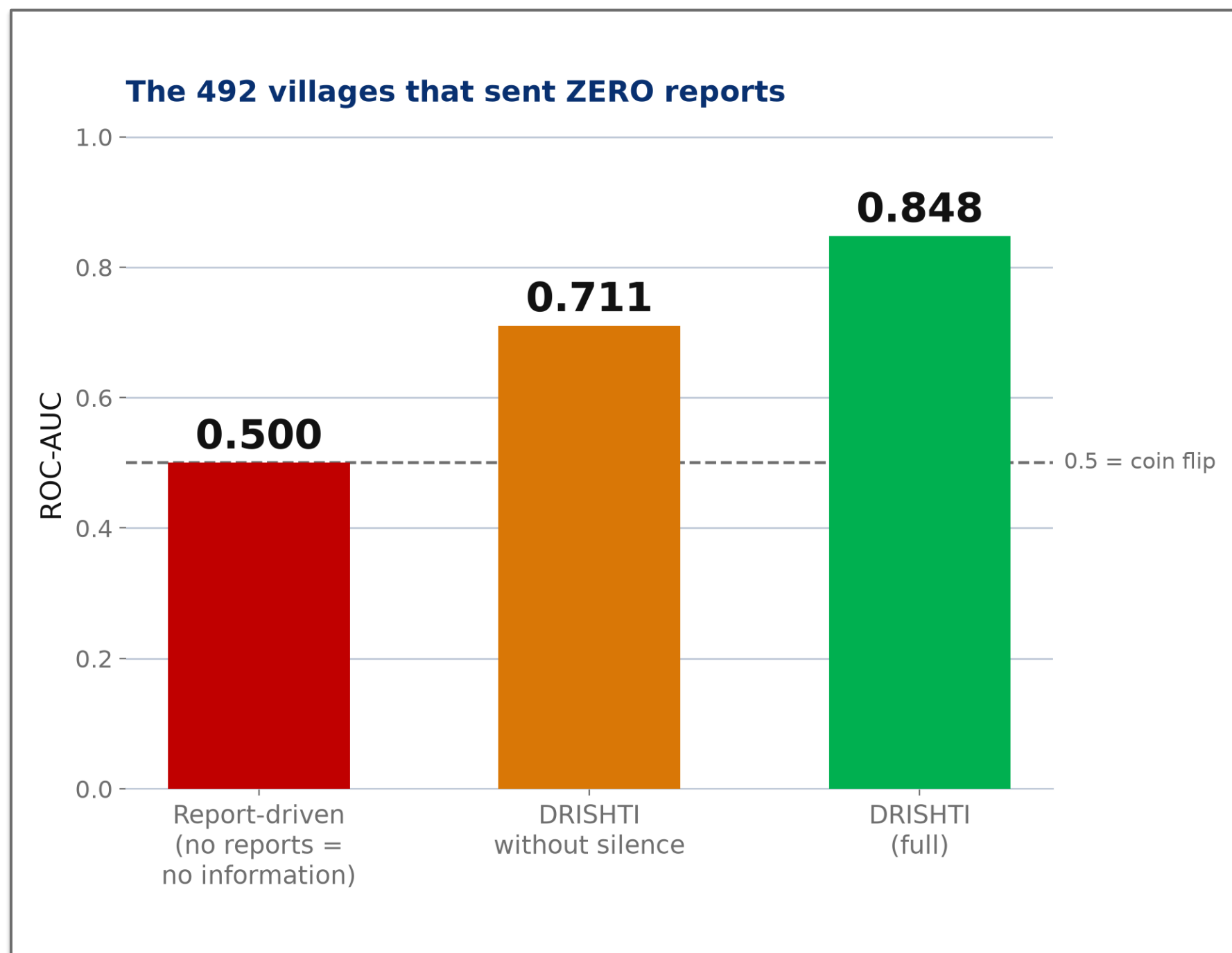


Fig. 2 Detection accuracy on the 492 settlements emitting no reports.

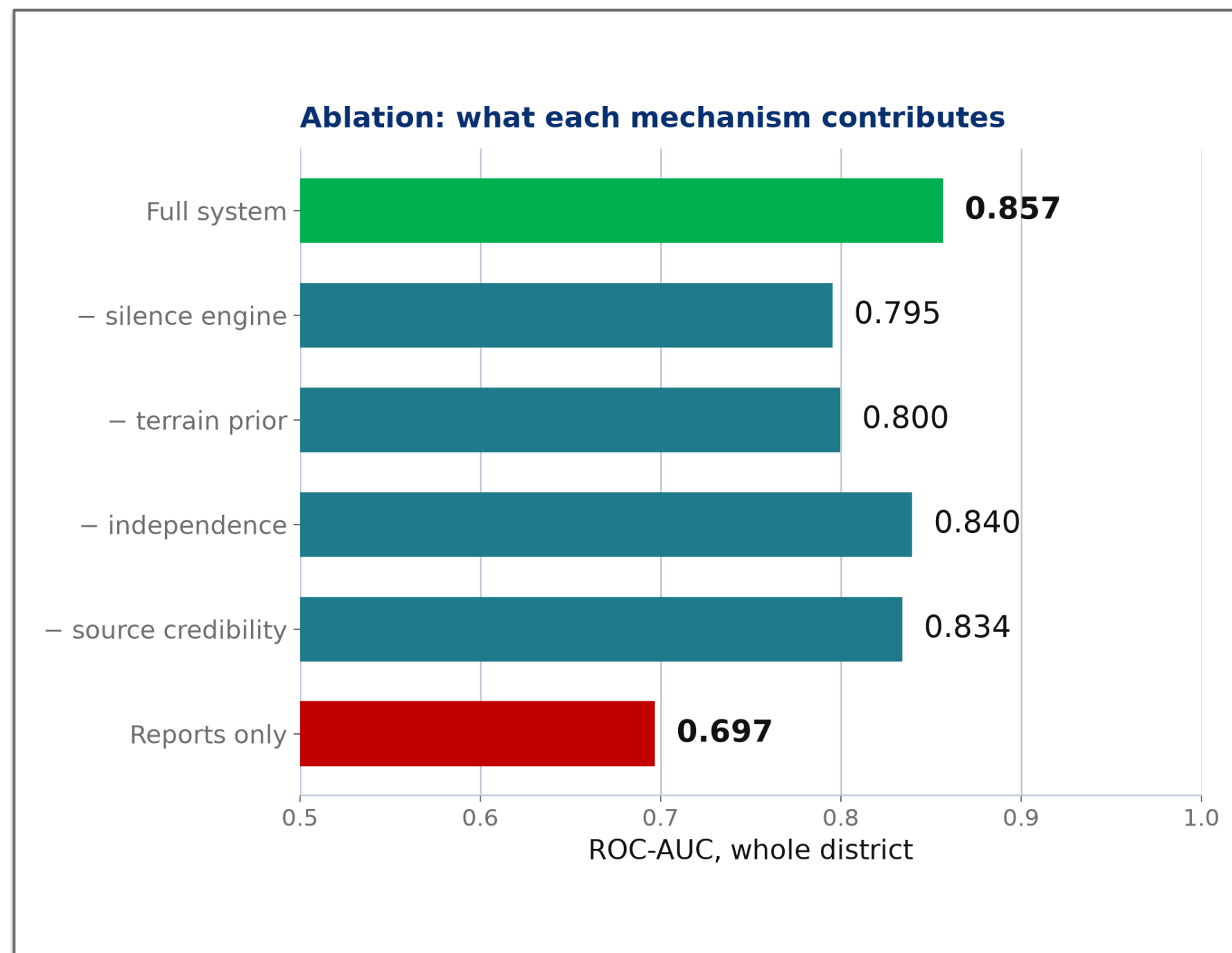


Fig. 3 Ablation: contribution of each mechanism to district-wide AUC.

Case study — Agastmuni, Rudraprayag (real settlement, pop. 19,758)

Reports expected in 24 h	418
Reports actually received	1
Cell tower	dead 23.9 h
Ground truth	CATASTROPHIC
Casualties / trapped	2,534 / 7,795

RANKED BY
REPORT VOLUME

#525

of 1,353
help never arrives

RANKED BY
DRISHTI

#1

of 1,353
first vehicle sent

Fig. 4 Case study. A real settlement of 19,758 people that emitted one report where 418 were expected.

Confusion matrix (n = 1102)

	Predicted			
	Intact	Minor	Major	Catas.
Intact	194	111	39	8
Minor	145	150	65	15
Major	20	67	79	87
Catas.	0	17	57	48

Calibrated performance

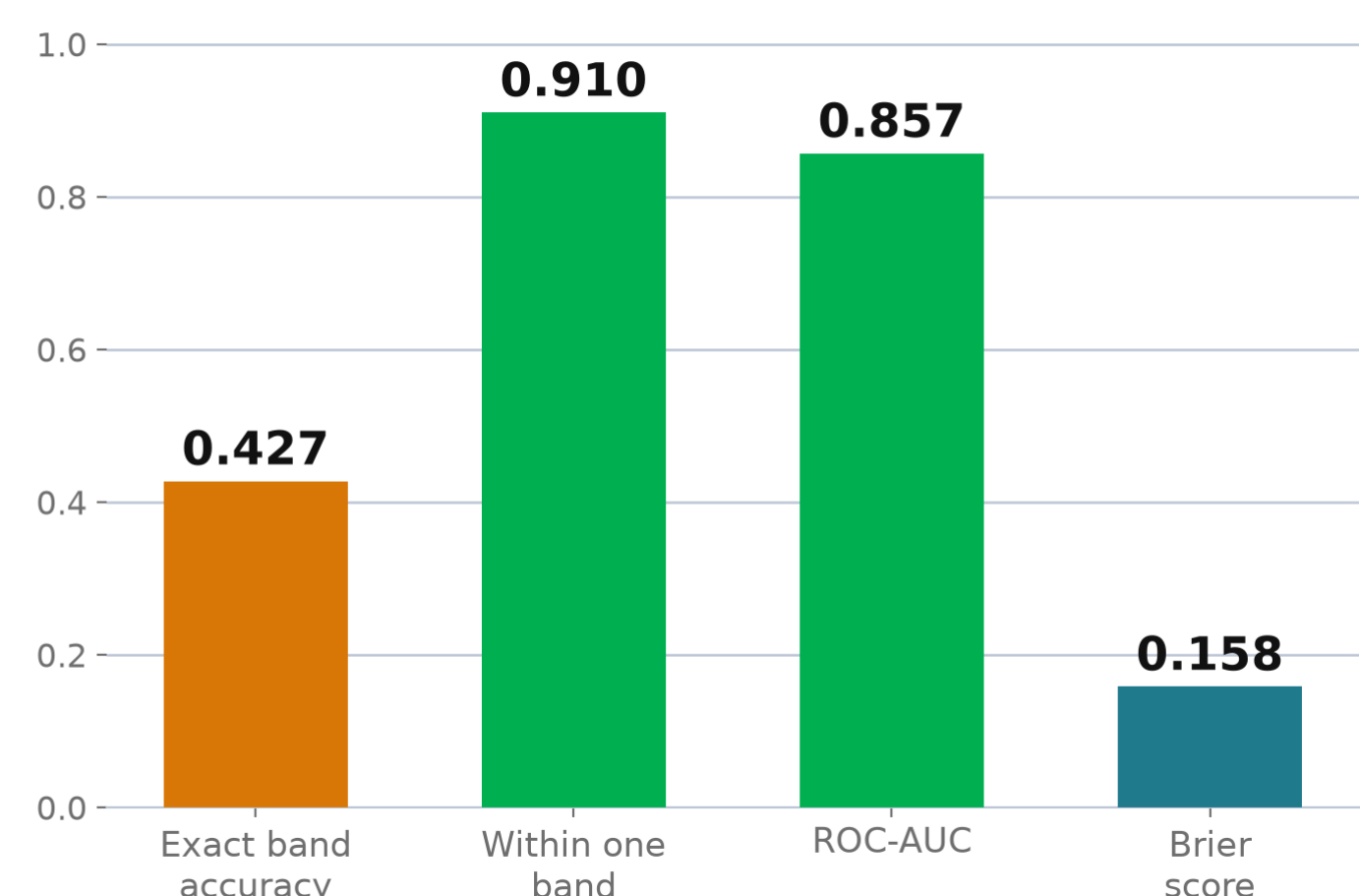


Fig. 5 Confusion matrix over 1,102 settlements and calibrated performance after temperature scaling.

CONCLUSIONS

1. Report volume is a hill, not a slope. Ranking by it systematically de-prioritises the highest-mortality settlements.
2. Absence of expected communication, fused with infrastructure liveness and terrain exposure, is a usable and calibrated signal of catastrophe.
3. On settlements that never report, report-driven triage is equivalent to a coin flip (AUC 0.500); DRISHTI achieves 0.848.
4. The system reaches 6.0x more affected people within the golden 24 hours using the same vehicles, and wastes fewer deployments.
5. Because it runs offline on free and open data, it is deployable where connectivity and budget are the binding constraints — the conditions that define a disaster.

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LIVE DASHBOARD
realgauravvyas.github.io/drishiti



SOURCE CODE
github.com/realgauravvyas/drishiti

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